REVIEW

The spectrum of inflammatory responses

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Inflammation is an integral part of animal biology. It provides critical protection from adverse environmental factors by enforcing the defense of homeostasis and the functional and structural integrity of tissues and organs. Recent advances have uncovered a broad range of biological processes that involve inflammatory control, calling for a renewed framework of inflammation beyond its classical roles in defense from infection and injury. In this Review, new perspectives on inflammation biology are discussed and new research directions are suggested to address the fundamental gaps in our current understanding.

Inflammation is a biological phenomenon that defies a simple definition. Depending on the context, inflammation can be thought of in terms of a response, a process, or a state of the system. Each of these notions emphasizes different aspects of inflammation: as a response to perturbation (e.g., infection or injury); as a multistep process of mobilizing defensive mechanisms to eliminate the source of perturbation; or as an altered state of the system that can be either protective or pathological. A renewed interest in inflammation biology is driven by the realization that inflammation is associated with almost every human disease (*1*). In addition, a growing body of evidence indicates that the cellular and molecular mediators of inflammation are involved in a surprisingly broad range of biological processes, including tissue remodeling, metabolism, thermogenesis, and function of the nervous system (including behavior) (*2*). This makes a formal definition of inflammation challenging and existing definitions arbitrary: If some biological process involves cytokines and myeloid or lymphoid cells, does it mean it is an inflammatory process? And if so, what makes it different from non-inflammatory processes?

One way to conceptualize this conundrum is to think of inflammation on a spectrum. At one extreme is acute inflammation caused by infection or injury. This is the most familiar type of inflammation orchestrated by local changes in microvasculature, resulting in inflammatory exudate (fluids, proteins, and cells in the extravascular space) that gives rise to the cardinal signs of inflammation (redness, swelling, heat, and pain). At the other end of the spectrum, cells and molecules known for their role in acute inflammation participate in normal homeostatic processes in the absence of any perturbations (*3*). Thus, the so-called "inflammatory" signals (cytokines, chemokines, eicosanoids, and bioactive amines) and cell types (macrophages, neutrophils, eosinophils, mast cells, and lymphocytes) are involved in both acute inflammation and in a variety of homeostatic processes. This suggests that there is some functional continuity in the operation of these signals and cell types. For example, both homeostatic and inflammatory functions of macrophages rely on phagocytosis, and both homeostatic and inflammatory functions of histamine and prostaglandins involve their effect on vascular smooth muscle and endothelium. The difference between "homeostatic" and "inflammatory" modes of action is sometimes quantitative, being related to the magnitude of the response. In other cases, it is qualitative in that it involves distinct gene expression programs and effector responses (compare, for example, macrophage phagocytosis of apoptotic cells versus bacteria). Between these extremes, the same inflammatory signals and cell types that are involved in acute inflammation and control of homeostasis participate in a variety of physiological responses to hostile environmental factors such as cold and starvation.

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Given the diversity of biological processes involving inflammatory signals and cells, the traditional view of inflammation as a response to infection or tissue injury appears somewhat narrow and incomplete because inflammation can clearly be present in the absence of these triggers. This is the case with inflammation associated with obesity, aging, cancer, and even sleep deprivation (*3*–*6*). One common denominator of various causes of inflammation is a deviation of a biological system from its "normal." homeostatic state in response to some perturbation (*7*–*9*). The causes of these deviations can be either environmental or internal; however, a formal definition of these inflammation-inducing perturbations is still lacking.

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Two modes of recognition of inflammatory inducers

External and internal perturbations that trigger an inflammatory response can be detected by two distinct modes of recognition (*9*). For example, infection can cause tissue damage and tissue damage would cause an inflammatory response. Here, the response would occur after the fact, or after damage is done. This is a retrospective mode of induction. However, infection can also be detected through pattern recognition receptors. For example, bacterial cell wall components such as lipopolysaccharide (LPS) can be detected by Toll-like receptors (TLRs) to induce an inflammatory response even before the bacterial pathogen has had a chance to cause any tissue damage (indeed, even in the absence of a pathogen). This is a prospective mode of induction of inflammation. LPS serves as a reliable proxy for Gram-negative bacterial infection. Prospective recognition is obviously preferable because it allows for pre-emptive measures (here, engagement of antibacterial defenses) that can minimize the damage that the pathogen would ultimately cause. Other inducers of inflammation, including virulence factors, toxins, and allergens, can similarly be detected prospectively by detecting characteristic activities associated with their damaging effects on the host (*7*, *10*–*12*).

Although prospective recognition of a threat is preferable, it is only possible when the noxious stimulus is part of the natural environment in the evolutionary history of the organism. In addition to infection, the common natural hostile environmental factors include starvation, dehydration, predation, envenomation, hypoxia, hypothermia and hyperthermia, and exposure to toxins (*13*). These environmental insults are major factors of extrinsic mortality and, as such, they are important drivers of natural selection. A key outcome of this selective pressure is the evolution of specialized pre-emptive defenses that protect from the morbidity and mortality caused by common hostile environmental factors. Accordingly, whenever possible, specialized defense mechanisms are engaged by these environmental factors prospectively, before they can cause the actual problem. Such mechanisms include behavioral defenses (e.g., foraging for food and water, habitat selection, and fight-or-flight response) and physiological defenses (e.g., thermoregulation, osmolyte excretion, lipolysis, and ketogenesis, etc.). When these preventive responses to food or water scarcity, predators, cold, and heat are successful and sufficient, tissue and organ damage are avoided. Growing evidence indicates that inflammatory responses are involved in these physiological and behavioral defenses to promote adaptation to environmental stressors (*2*, *14*). In these cases, "physiological inflammation" occurs in the absence of tissue damage. Conversely, pathogens, venoms, and toxins are mostly detected once infection, envenomation, or ingestion have taken place, respectively. Even though prospective recognition is still engaged, avoiding tissue damage is nearly impossible. Consequently, the more familiar and better studied acute inflammatory response is a critical part of defenses against this class of threats.

Three types of perturbations leading to inflammation

What basic types of perturbations can a system experience? Both biological and engineered systems have three universal characteristics: The elements of the system are arranged and interconnected into a particular structure, the structure supports a specific function, and regulation of the function is performed according to some logic of the system such as growth, stability, or coordination. Any one of these characteristics can be affected by a perturbation, which means that, in general, a system can experience three types of problems: loss of regulation, loss of function, or loss of structure. These three characteristics have an obvious hierarchy in that structure is required for function and function is required for regulation. Therefore, loss of structure entails loss of function and loss of function implies loss of regulation.

In a biological system, all three types of problems can occur in response to different environmental or internal perturbations, and all three can elicit protective inflammatory responses (Fig. 1). These responses supplement, enhance, or otherwise modify the normal structural, functional, and regulatory features of the system to optimize defense from, or adaptation to, the corresponding disruptive perturbations. For this discussion, the term "inflammation" will be used in a broad sense as any process involving signals and cells known to orchestrate the more familiar acute inflammatory response.

Response to loss of structure

Loss of structure, or tissue damage, is a well-known cause of inflammation. Inflammation induced by loss of structure (sometimes called "type 1 inflammation") is characterized by an acute microvascular response, edema, and recruitment of neutrophils and monocytes (*15*). This type of inflammation is also caused by acute infections with microbial pathogens that can cause tissue damage (*16*). Currently, the best understood mechanism of initiation of inflammation is by pattern recognition of conserved molecular structures of microorganisms (*3*, *17*). This is a case of prospective recognition that occurs before pathogens cause tissue damage. Retrospective recognition of tissue damage is generally based on detection of loss of compartmentalization or sequestration. Thus, molecular cues that are normally sequestered, for example, inside the cells or their organelles, are released upon damage to initiate inflammation (*3*, *18*) (Fig. 1A).

Acute inflammatory responses to infection and tissue damage aim to eliminate the source of perturbation (e.g., pathogens) and to promote repair of lost structure. In addition to repair of the damaged tissues, inflammatory responses may supplement existing tissue structures with modifications that optimize defense from structural perturbations. For example, barrier epithelia undergo inflammatory remodeling to increase the abundance of mucus-producing goblet cells and to alter mucus structure and composition in the intestine (*19*) and airways (*20*). Other examples of inflammatory remodeling include tertiary lymph nodes and granulomas, both of which are locally generated at the sites of chronic inflammation (*21*, *22*). These examples illustrate the distinction between basal or "steady-state" structures and "inflammatory" structures: The inflammatory structures are not required for homeostatic function, they are induced on demand in response to acute or chronic perturbation, and they are products of inflammatory tissue remodeling (change in tissue composition and architecture) (Fig. 2A). Although inflammatory tissue remodeling aims to protect the tissues from further damage, persistent, exaggerated, or otherwise abnormal remodeling is one of the drivers of chronic inflammatory pathologies, as

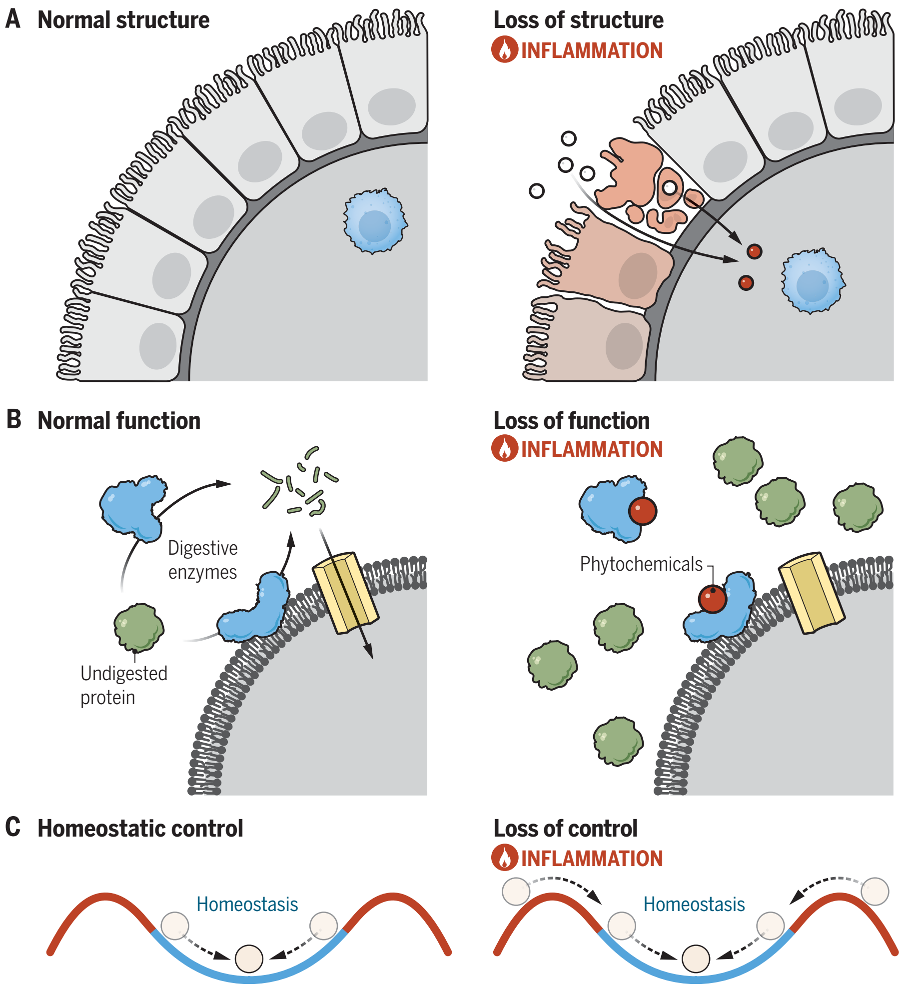

Fig. 1. Three types of perturbations that can lead to inflammation. (**A**) Loss of structure (tissue damage) results in desequestration of intracellular molecules and loss of tissue compartmentalization (e.g., loss of epithelial and endothelial integrity). Molecules released from damaged cells and organelles, as well as luminal components of the intestine or blood vessels, activate type 1 inflammatory responses. (**B**) Loss of function may activate type 2 inflammatory responses. For example, inhibition of digestive enzymes (blue) in the small intestine by phytochemicals (red ball) would lead to accumulation of undigested proteins (green) and may result in allergic inflammation. (**C**) Loss of homeostasis can produce a distinct type of inflammatory response, the physiologic inflammation that occurs in the absence of infection or injury. When homeostatic mechanisms are insufficient to maintain the stability of the system, physiologic inflammation enforces restoration of homeostasis state. Although tissue damage is a well-known inducer of inflammation, loss of function and loss of regulation are hypothetical scenarios that require further investigation.

seen in airway remodeling in asthma and in a variety of fibrotic diseases (*23*).

Response to loss of function

Transient or permanent loss of function can happen for a variety of reasons, including as a secondary consequence of loss of structure. However, direct loss of function is commonly caused by noxious substances, including environmental toxins, xenobiotics, and some venom components (Fig. 1B). One important example is phytochemicals that plants produce as a defense against herbivory. Some phytochemicals are designed to interfere with digestive functions (*24*, *25*) and others compromise epithelial barrier functions (*26*). Large airborne particles (e.g., pollen and dust) likely interfere with ciliary function of the airway tract. These types of threats that cause loss of function are characteristic of allergens and xenobiotics (*27*). They elicit type 2 inflammation, which is characterized by eosinophil recruitment, alternative activation of macrophages, and the production of inflammatory mediators by mast cells (*28*). This type of inflammation is also caused by parasitic worms (*29*, *30*), raising the possibility that at least some aspects of recognition of parasite infections might be based on detection of some functional perturbations.

What is the purpose of inflammatory responses to loss of function? Like inflammation induced by loss of structure, inflammation elicited by loss of function has two goals. One goal is to eliminate the agents causing functional alterations, which is achieved by promoting the expulsion of noxious agents or parasites by activating peristalsis, diarrhea, vomiting, itching, sneezing, and coughing. Another goal is to complement a basal, or steady-state, mode of function with an "inflammatory" mode of function. An example of "functional complementation" is the inflammatory matrix degradation mediated by macrophage- and neutrophil-derived proteases, which complements homeostatic turnover of extracellular matrix. Another example is inflammation-induced peristalsis, which is driven by histamine and serotonin produced by mast cells and enterochromaffin cells to promote the expulsion of toxins from the intestinal lumen (*27*). This inflammatory function complements homeostatic peristalsis (regulated by mechanosensing of luminal material), which would be insufficient to provide defense from toxins (Fig. 2B).

Virulence factors, toxins, and allergens can be detected prospectively, based on their unique characteristics such as biochemical activities, or retrospectively because they cause loss of function or, in extreme cases, loss of structure that may follow if other forms of recognition are ineffective or not available. Key questions for future studies are: Which functions are monitored such that their loss results in an inflammatory response? And which basal functions can and which cannot be complemented by inflammatory functions?

Response to loss of regulation

Finally, loss of regulation can promote yet another type of inflammation, which may be best described as "physiological inflammation," i.e., inflammation involved in the regulation of physiological processes. This type of inflammation can occur in the absence of infection, tissue damage, or exposure to noxious substances. Instead, it takes place in response to other hostile environmental factors (e.g., cold, starvation, predation) that threaten organismal

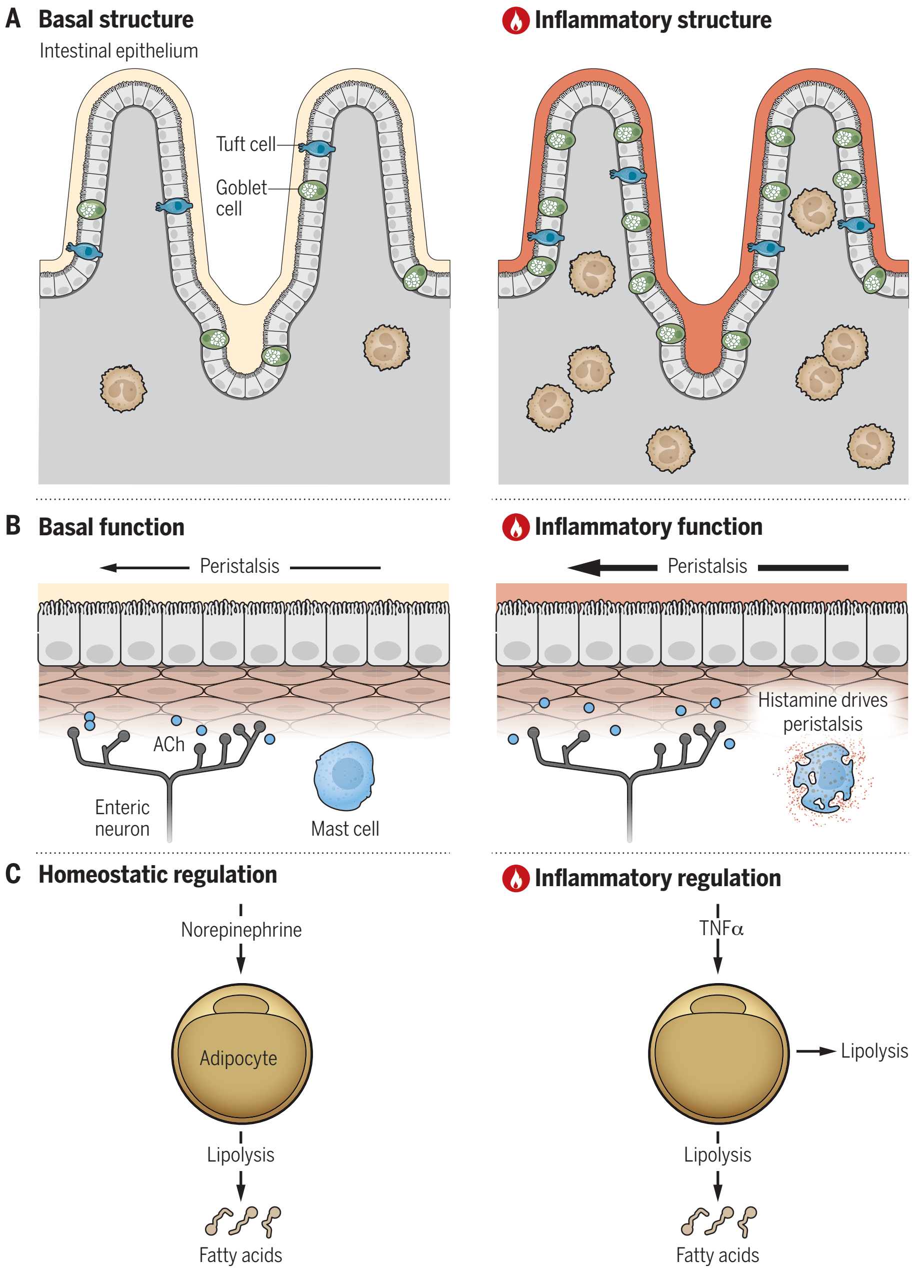

Fig. 2. Adaptive arms of inflammatory responses. In addition to the elimination of causative agents, inflammatory responses promote adaptation to structural, functional, or regulatory perturbations. (**A**) Inflammation induced by loss of structure promotes tissue remodeling to generate alternative ("inflammatory") structures. In the example shown here, the basal structure of the intestinal epithelium is remodeled because of goblet cell (green) and tuft cell (blue) hyperplasia and increased mucus production. The resulting tissue structure is more damage resistant. (**B**) Certain functions can operate in normal and inflammatory mode. Here, normal intestinal peristalsis, which is controlled by acetylcholine produced by enteric neurons, is contrasted with inflammatory peristalsis, which is controlled by histamine produced by activated mast cells. The inflammatory mode helps to expel parasitic worms and allergens. (**C**) Under normal conditions, lipolysis is regulated by norepinephrine produced by the sympathetic nervous system. Under inflammatory conditions, lipolysis is induced by inflammatory cytokines, including TNF.

homeostasis. Examples of physiological inflammation include regulation of thermogenesis (*2*, *31–33*), control of metabolic homeostasis (*34*), and the fight-or-flight response (*14*). Thus, physiological inflammation can be viewed as defense of homeostasis.

Homeostatic regulation operates based on negative feedback mechanisms that correct deviations of the system's state variables from the desired range, or setpoint values. When the deviations are large enough, homeostatic mechanisms can be insufficient to maintain a system's stability. In these cases, inflammatory signals are engaged to complement the homeostatic regulation and enforce the return to homeostasis (Fig. 1C). This is exemplified by control of body temperature, where cold exposure activates homeostatic response through activation of the sympathetic nervous system, with adrenergic signaling promoting thermogenesis by brown fat (*35*). With more severe or prolonged cold challenge, however, sympathetic activation of preexisting brown adipocytes is insufficient and additional mechanisms involving inflammatory pathways are engaged. Specifically, cold challenge activates adipose tissue γδ T cells to secrete tumor necrosis factor-α (TNFα) and interleukin-17A (IL-17A), leading to the production of IL-33 in adipose stromal cells (*36*). Activation of ILC2 cells by IL-33 results in the recruitment of eosinophils, leading to IL-4–driven differentiation of brown and beige adipocytes (*31–33*). Increased beiging and browning of adipose tissue then leads to increased thermogenesis. Similar mechanisms contribute to the maintenance of energy homeostasis under conditions of fasting and exercise (*34*). We can again see how inflammatory regulation complements homeostatic control; for example, lipolysis is normally regulated by adrenergic signaling and growth hormone, but under inflammatory conditions, lipolysis is induced by cytokines such as TNFα (Fig. 2C).

Another way that inflammation contributes to homeostatic control is by changing homeostatic setpoints in the hypothalamus. This is best known under conditions of acute inflammation caused by infection and resulting in changes in physiological state known as "sickness behaviors," which include changes in homeostatic setpoints for body temperature, appetite, sleep, etc. (*37*, *38*). One might hypothesize, however, that milder versions of setpoint changes may occur in response to loss of homeostatic regulation under a much broader set of adverse conditions, especially when they are chronic. For example, weight gain, hypertension, anxiety, and depression are all consequences of changes in the corresponding homeostatic setpoints. It would be important to investigate whether inflammation is the primary cause of setpoint changes in these settings. Loss of cellular, tissue, and organismal homeostasis is one of the features of aging and declined health (*6*, *39*). It is therefore likely that aging-associated inflammation may stem, at least in part, from loss of homeostatic regulation. How loss of regulation initiates inflammation is not well understood, perhaps because it has not yet been systematically studied. It will be an important direction for future research.

All three types of system perturbations, loss of structure, loss of function, and loss of regulation, can elicit inflammatory responses (Fig. 1). These responses have different modalities characterized by involvement of different cellular and molecular components and different magnitudes of the responses. All three types of inflammatory responses have two goals: One is the elimination of whatever caused the perturbation and another is adaptation to structural, functional, or regulatory perturbations. Achieving these goals can be complementary to some and antagonistic to other aspects of physiology. For example, increased mucus production supports the barrier function but interferes with epithelial transport function. Defining which functions are complemented and which are antagonized will help us better understand the logic of the inflammatory responses.

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Priorities of inflammatory responses

Whether inflammation is complementary or antagonistic to homeostasis depends on the relative priorities of different functions under given conditions. If deviation of homeostasis is the primary problem, then the inflammatory response complements homeostasis. If loss of homeostasis is secondary to a more serious problem (e.g., infection or tissue damage), then inflammatory response to infection or damage will have higher priority over restoration of homeostasis. Moreover, any homeostatic controls or any functions in general that are incompatible with the higher-priority inflammatory response are strategically suppressed by inflammatory signals. For example, homeostatic control of metabolic fuel allocation is overridden by inflammatory cytokines. Mechanistically, this is due in part to induction of insulin resistance in skeletal muscle, adipose tissue, and liver by TNFα, IL-1, and IL-6 (*40*). Here, inflammatory control is antagonistic to homeostatic regulation to allow fuel distribution to support higher-priority functions (defense from infection). In addition, the logic of the inflammatory response (what is activated and what is suppressed) reflects another type of hierarchy: Because one problem can lead to another (e.g., infection leads to tissue damage, which leads to loss of function, which leads to loss of regulation), primary problems are typically addressed first, before attending to secondary problems. Indeed, an inflammatory response clears the infection first, before attempting to repair the damage caused by the infection. This is also reflected in the succession from type 1 (induced by loss of structure) to type 2 (induced by loss of function) inflammatory responses to tissue damage (*29*). In general, responses to structural, functional, and regulatory perturbations operate in the order of decreasing priority because of the natural hierarchical relations among structure, function, and regulation. Accordingly, structural damage must be addressed first, before function and, ultimately, regulation can be restored.

Inflammatory circuits

Inflammation is orchestrated by dozens of molecular mediators and involves multiple cell types. Although it is known that there are qualitatively distinct types of inflammation elicited by different stimuli, particularly in the context of infections (e.g., type 1 and type 2 inflammation), the general framework for classification of inflammation is still lacking. This problem is challenging because most types of inflammation involve overlapping sets of cytokines and effector cells. Perhaps dissecting the causes of inflammation (loss of regulation, function, or structure) might yield a better classification framework.

Regardless of qualitative characteristics, all inflammatory responses are organized around four universal components that define an inflammatory circuit (Fig. 3). These include the molecular cues that trigger the response, sensors that detect them, inflammatory signals produced by the sensors, and various targets of inflammatory signals. There are two types of targets for inflammatory signals: First, there are effectors, cell types that are directly involved in elimination of the inflammatory triggers. Inflammatory effectors are composed of various specialized cell types (macrophages, monocytes, neutrophils, eosinophils, mast cells, and basophils), as well as barrier epithelial cells (goblet and Paneth cells). Inflammatory signals induce activation, recruitment, and differentiation of the effector cells. This part of the inflammatory circuit operates in a negative feedback fashion: The trigger of the response (e.g., a pathogen) is eliminated by the response effectors.

In addition to effector cells, however, inflammatory signals act on most other tissues and organs that do not directly participate in the elimination of pathogens. These tissues and organs are affected by inflammatory signals in a prescribed way: Whatever functions the target tissues normally perform (e.g., secretory, contractile, filtering or barrier function) are either up-regulated or down-regulated by different inflammatory signals. In other words, inflammatory signals control the same functions that different tissues normally perform under control of homeostatic signals (*8*, *9*). In principle, depending on what the problem is (loss of regulation, function, or structure), inflammatory signals can change these functions either in the same "direction" as homeostatic signals (complementary function of inflammation) or they can be antagonistic to homeostatic signals. For example, vasodilation can be increased both by adenosine due to local hypoxia (homeostatic control) and by inflammatory signals, including prostaglandins, to increase blood supply to the affected tissue. In this example, homeostatic control is insufficient and inflammatory signal acts to enforce homeostasis. Conversely, when inflammation is induced by loss of function or structure (as in infection or tissue damage), attending to these problems has higher priority than maintaining homeostatic regulation. In such cases, inflammatory signals often operate in a manner antagonistic to homeostatic signals.

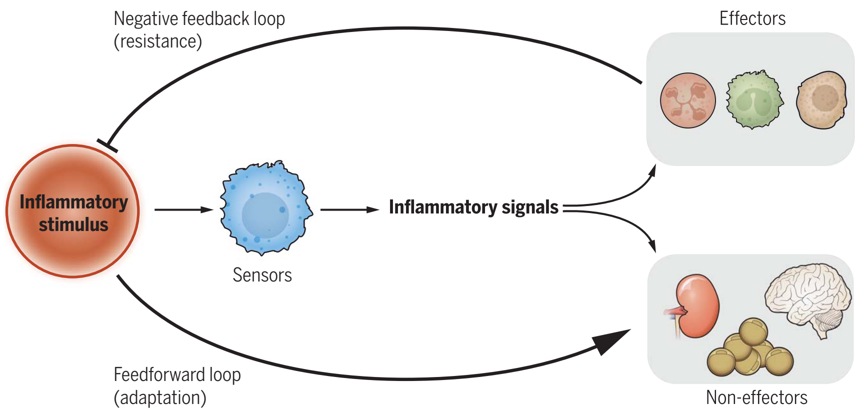

Fig. 3. Inflammatory circuit. Inflammatory pathways consist of an inflammatory stimulus, sensors, and inflammatory signals that act on two types of targets. One type is the inflammatory effectors, such as neutrophils, monocytes, and eosinophils. Effectors eliminate the cause of inflammation. This is a negative feedback (resistance) part of the circuit. Another type of target are tissues and organs that change their functional state to adapt to the perturbation and to support the effector response. This is a feedforward (adaptation/tolerance) part of the circuit.

Overall, inflammatory signals have two types of targets: effectors, which eliminate the cause of the response, and "non-effectors," tissues and organs that do not directly participate in eliminating the threat but still undergo specific changes directed by inflammatory signals. The effect of inflammation on these non-effector targets promotes adaptation to the inflammatory state of the organism, inducing the state of tolerance to infection (*41*, *42*). This includes down-regulation of incompatible lower-priority functions, redirection of metabolic resources to the effectors, and generally providing permissive physiological state for the defensive task at hand. Because activation of effectors leads to elimination of the cause of inflammation, this part of the circuit operates in a negative feedback fashion. This is the resistance part of the response. The part of the circuit that involves non-effector tissue and organs operates in a feedforward manner (i.e., the response does not affect its own cause but rather the downstream target). This is the adaptation (tolerance) part of the response (Fig. 3). In addition to the bona fide inflammatory signals, which target both effectors and non-effectors, the adaptation circuits can be controlled by specialized signals such as GDF15, which is induced under a variety of noxious conditions, including infections and toxins, and promotes adaptive physiological changes (*43*, *44*).

These two arms of the response, resistance and adaptation, are the common theme in inflammation biology. When the resistance arm is dominant, the response has all the characteristics of acute inflammation. When resistance is unsuccessful and the source of perturbation is persistent, then the adaptation arm of the response becomes dominant, and inflammation becomes chronic. Thus, acute and chronic inflammation, although differing in kinetics, are actually qualitatively different types of responses.

Both resistance and adaptation parts of the inflammatory circuit can result in pathological outcomes when these responses are excessive. The resistance arm of the response can lead to collateral tissue damage and other forms of immunopathology, where inflammatory effectors negatively affect tissues and organs. The adaptation arm of the response can result in a different class of pathologies caused by excessive changes in functional states of non-effector target tissues and organs.

Regulation of inflammation

Successful inflammatory responses eliminate the cause of inflammation (e.g., pathogens). This should in principle terminate the response. However, it is well known that elimination of the pathogen is not in itself sufficient to control inflammation. In the absence of negative regulators, the inflammatory response is always overwhelmingly excessive to the point of being pathological. Studies of the regulation of inflammatory responses have therefore focused on the mechanisms controlling the response magnitude and duration. Specifically, a major focus has been on the regulation of inflammatory signal production and the activity of effector cells. The anti-inflammatory signals that control the magnitude of inflammation include several well-studied examples, including IL-10, transforming growth factor-β (TGF-β), and glucocorticoids (*45*–*47*). In addition, there is a plethora of negative regulators of inflammatory signaling pathways that operate in a negative feedback fashion (i.e., they are inducible by the inflammatory signals). These include suppressor of cytokine signaling (SOCS) proteins, negative regulators of Janus kinase–signal transducer and activator of transcription (JAK-STAT) signaling, and A20, a negative regulator of nuclear factor-κB (NF-κB) signaling (*48*, *49*). These signaling regulators control the magnitude and duration of the downstream response, such as the production of various cytokines. Anti-inflammatory regulators are expressed as a function of the magnitude of the response they regulate, not as a function of the output of the response (e.g., pathogen clearance). This is in contrast to signals that control resolution and repair, such as lipoxins and resolvins, which are produced as a function of response output, i.e., upon pathogen clearance (*50*). Indeed, resolution and repair would be futile and counterproductive if engaged before the wound is sterilized.

The magnitude and duration of the inflammatory response depend on the level and persistence of a threat (e.g., pathogen load). In addition, collateral damage caused by the inflammatory response must be factored in to compute the optimal strength of the response. How this is achieved is unclear, but there are two broad possibilities. One is that the strength of inflammatory response is a function of two variables, the strength of the input (e.g., pathogen load) and the cost of the response (e.g., collateral tissue damage). The latter would be expected to provide a negative feedback signal that reports on the degree of inflammatory tissue damage. According to this hypothesis, inflamed tissues may be producing anti-inflammatory signals that down-regulate the inflammatory response. One such signal is adenosine, which is produced by inflamed and stressed tissues and has anti-inflammatory effects when it signals through A2 receptors (*51*); additional signals with this property likely exist. The second, perhaps less likely, scenario is that the degree of inflammatory damage does not feed back on the strength of the response and the latter is simply calibrated over evolutionary time to provide the best response on average. These possibilities are experimentally testable and should yield important insights into the control of inflammation.

Given that optimal response is achieved by maximal effect (e.g., pathogen clearance) with minimal cost (e.g., collateral damage), is regulation of the magnitude of the response the only way to control inflammation? The problem with controlling the magnitude alone is that the cost of the response may constrain the strength of the response. For example, suppose that clearance of a given infection requires the production of 100 arbitrary units of TNFα, but that much TNFα is lethal to the host organism. In this scenario, there is no optimal strength of the response because any response would be either too weak (i.e., cannot clear the infection) or too strong (i.e., cannot survive the inflammatory damage). What are the possible solutions to this trade-off?

One way around this constraint is through differential responsiveness of target tissues to inflammatory signals so that vital tissues and organs most vulnerable to inflammatory damage have lower responsiveness to inflammatory signals (*41*). Another possible strategy is the reversal of the inflammatory state of tissues back to their original, "homeostatic" state. The signals with such effects would not affect the magnitude of the inflammatory response but would minimize its cost. Such signals could be referred to as counter-inflammatory signals

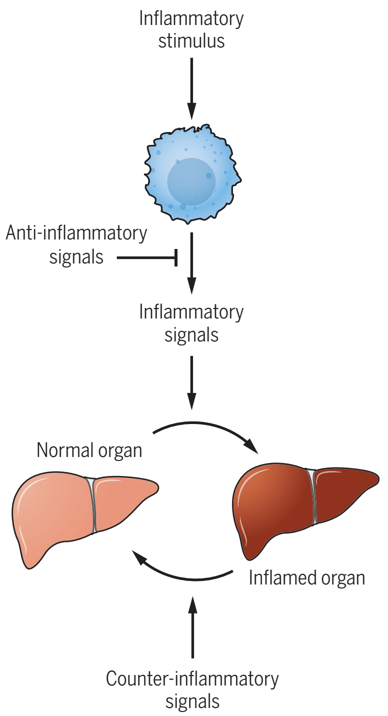

Fig. 4. Anti-inflammatory and counter-inflammatory signals. Anti-inflammatory signals control the magnitude of the inflammatory response. Counter-inflammatory signals reverse the effects of inflammatory signals on target tissues and organs that are not effectors of inflammation.

to distinguish them from anti-inflammatory signals that control the magnitude of the response (Fig. 4). Because deviation from the normal state is generally corrected by homeostatic signals, it is likely that these signals perform the function of countering the effect of inflammatory signals. One example of this is the effect of epinephrine on bronchial smooth muscles: Histamine and leukotrienes induce bronchoconstriction during an inflammatory response, whereas epinephrine counters that effect by inducing bronchorelaxation to return to homeostasis. In addition to homeostatic signals, however, more dedicated counter-inflammatory signals may exist that restore the homeostatic state of the tissues affected by inflammation. Likely sources of such specialized signals are T regulatory cells, which not only control the magnitude of inflammation but also tissue homeostasis (*52*). Signals involved in resolution (lipoxins and resolvins) (*50*, *53*) also likely have counter-inflammatory effects (in addition to their roles in terminating inflammatory responses). It should be noted that anti-inflammatory and counter-inflammatory effects are distinct modes of control of inflammation; the former controls the magnitude of the response, whereas the latter controls the degree of responsiveness of target tissues. As such, these effects do not necessarily have to be performed by distinct signals. For example, glucocorticoids, adenosine, and TGF-β all likely have both anti-inflammatory (i.e., acting on sensor and effector cells) and counter-inflammatory (i.e., acting on all other target tissues) modes of regulating inflammation.

Although existing anti-inflammatory drugs provide the desired therapeutic benefits, their common side effect is increased susceptibility to infections. Activating the counter-inflammatory pathways instead should help to restore tissue and organ homeostasis without compromising defense functions. In effect, such drugs would promote tolerance to inflammatory damage. Investigation of counter-inflammatory pathways will be an important direction for future research and for the development of a new class of therapeutics.

Conclusions

Inflammation as a defense system is essential for survival, but it is also inextricably associated with pathology and almost every known disease. Because infection- and injury-induced inflammation are the most prominent and most studied forms of the response, we may have skewed our understanding of inflammation according to these extreme conditions. Here, the case is made that inflammation has a far more fundamental role in physiology and that many important functions of inflammation have not yet been discovered. This includes the role of inflammation in monitoring, protecting, and enforcing the homeostatic, functional, and structural integrity of tissues and organs. Systematic studies of these aspects of inflammation biology should yield deeper understanding of the biology and pathophysiology of diseases.

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ACKNOWLEDGMENTS

Funding: This work was supported by the Howard Hughes Medical Institute, the Blavatnik Family Foundation, the Food Allergy Science Initiative, and the National Institutes of Health (grant AI144152-01). **Competing interests:** The author declares no competing interests.

10.1126/science.abi5200